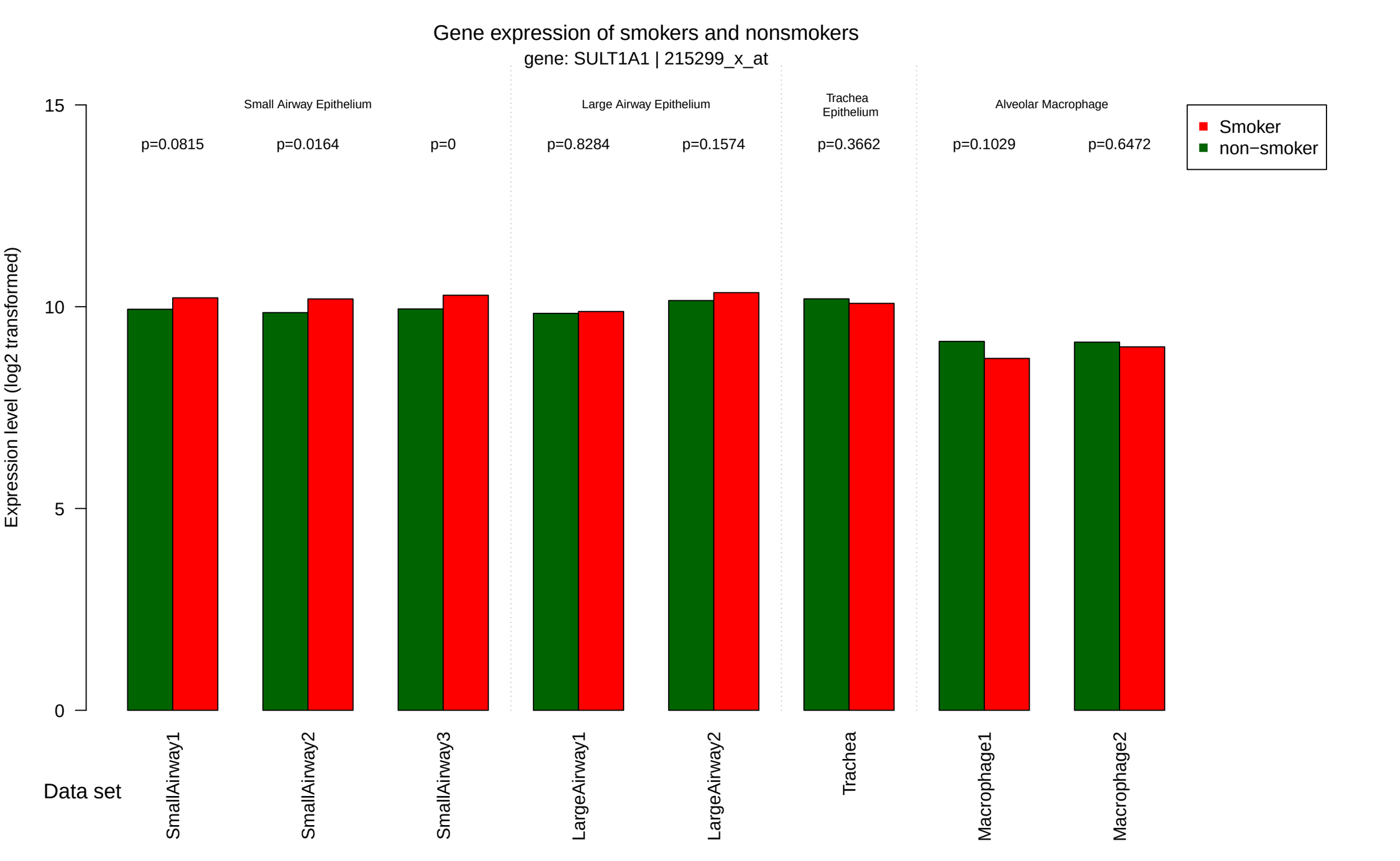
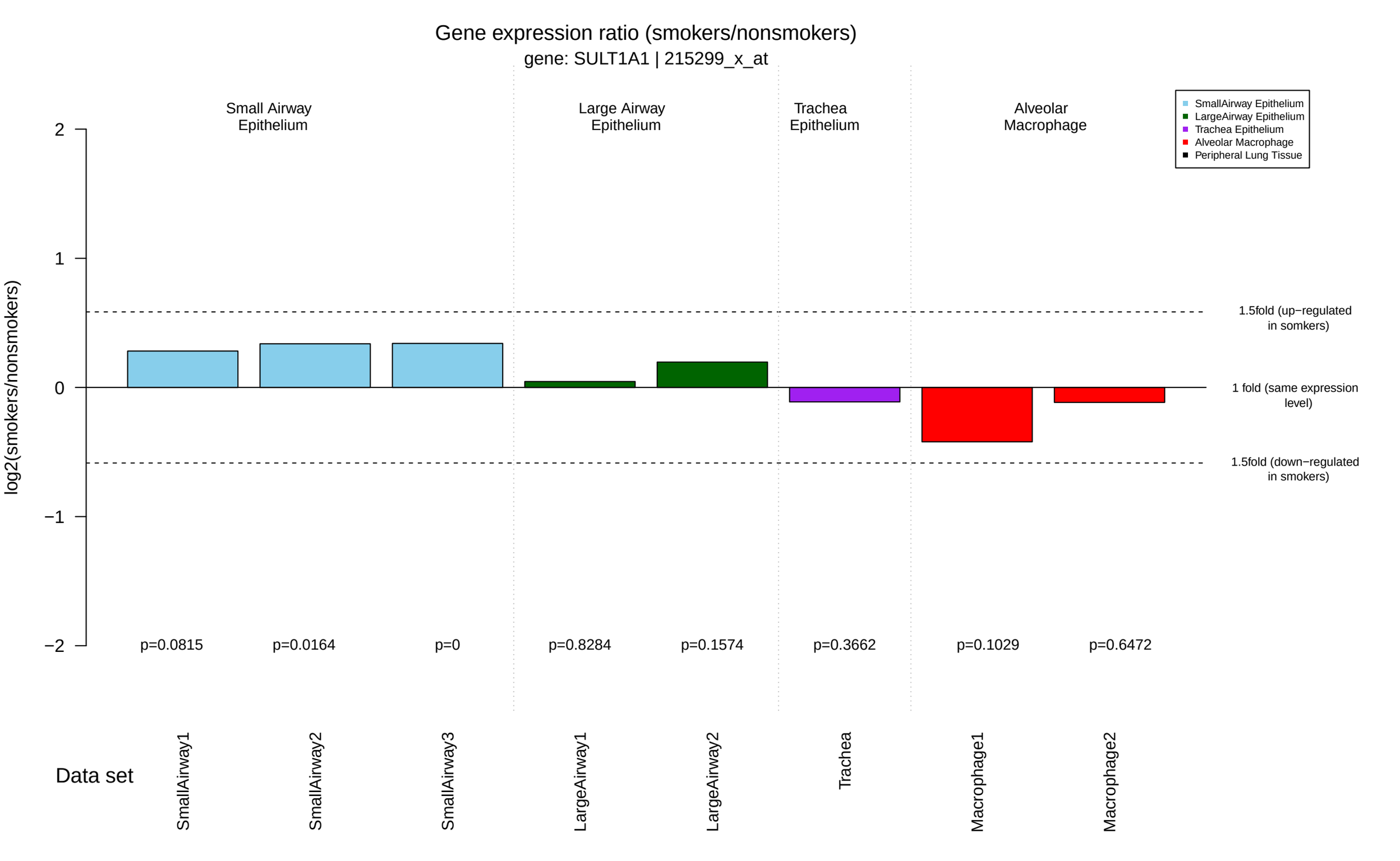
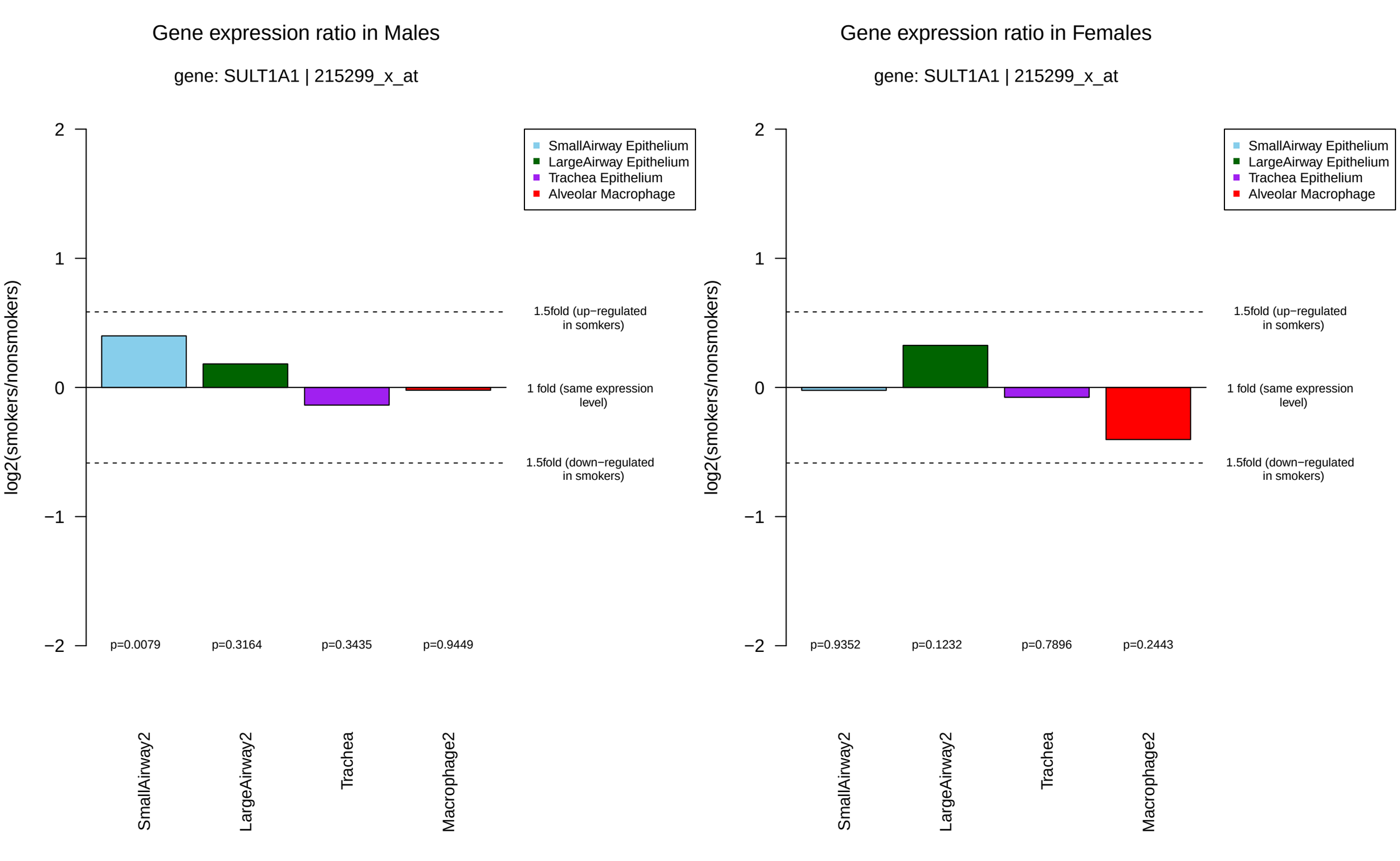
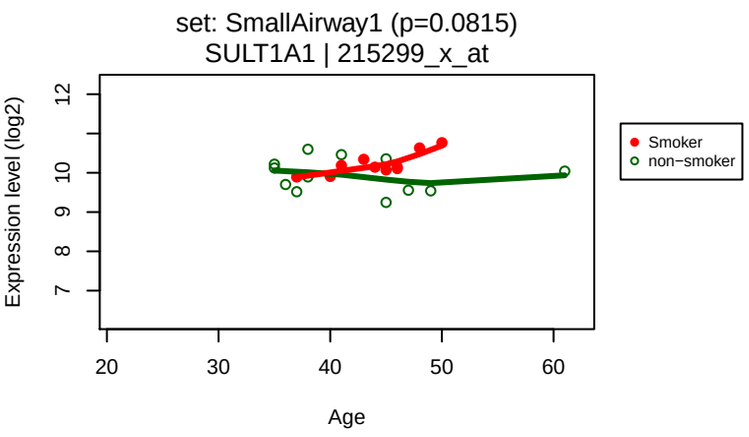
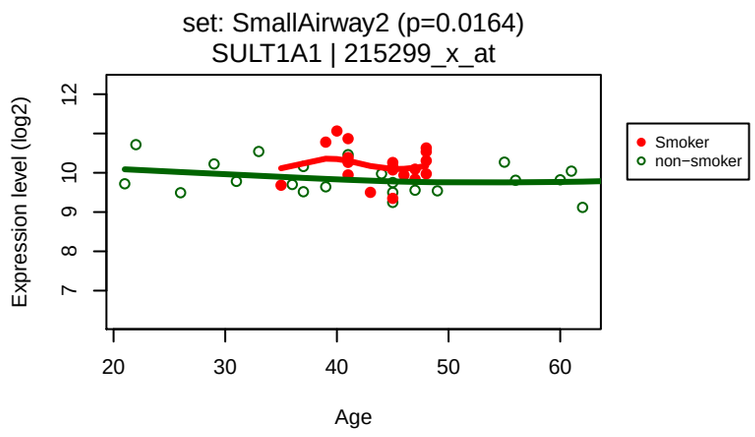
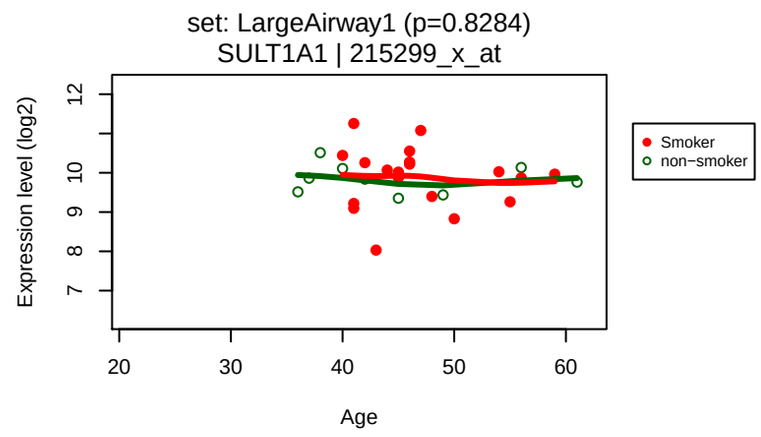

Gene expression of smokers and nonsmokers
gene: SULT1A1 | 215299_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0815
p=0.0164
p=0
p=0.8284
p=0.1574
p=0.3662
p=0.1029
p=0.6472
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

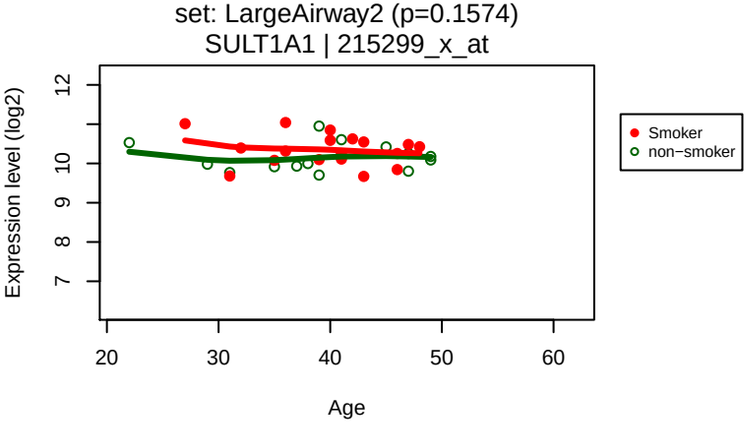

Gene expression ratio (smokers/nonsmokers)
gene: SULT1A1 | 215299_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0815
p=0.0164
p=0
p=0.8284
p=0.1574
p=0.3662
p=0.1029
p=0.6472
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

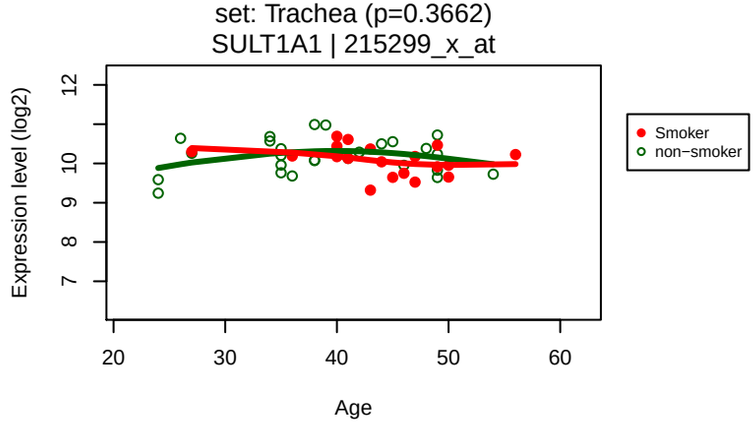

Gene expression ratio in Males
gene: SULT1A1 | 215299_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0079
p=0.3164
p=0.3435
p=0.9449
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SULT1A1 | 215299_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9352
p=0.1232
p=0.7896
p=0.2443
SmallAirway2
LargeAirway2
Trachea
Macrophage2

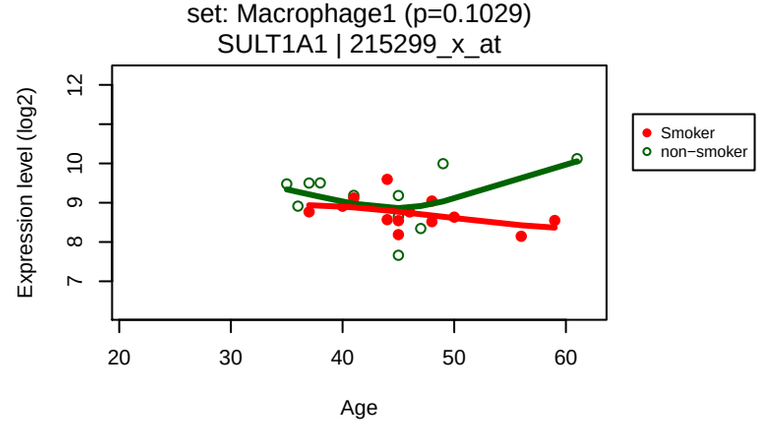

set: SmallAirway1 (p=0.0815)
SULT1A1 | 215299_x_at
Expression level (log2)
Age
Smoker
non−smoker

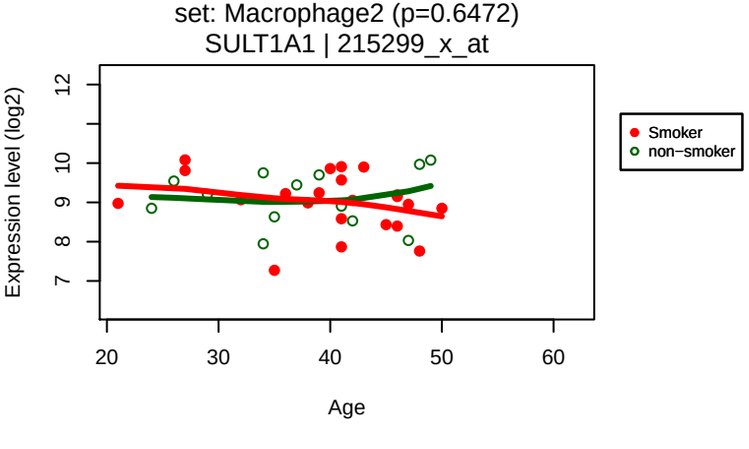

set: SmallAirway2 (p=0.0164)
SULT1A1 | 215299_x_at
Expression level (log2)
Age
Smoker
non−smoker

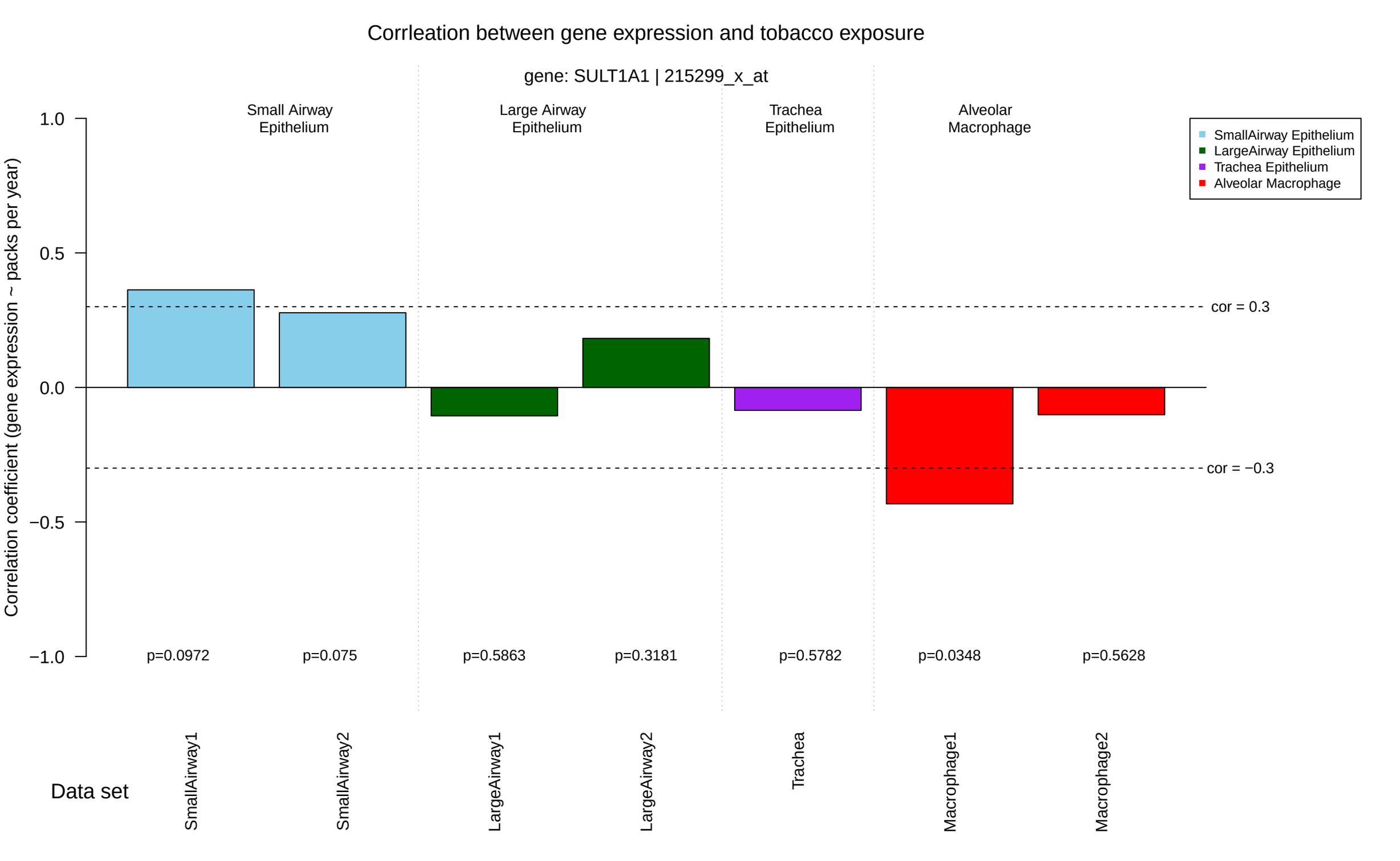

set: LargeAirway1 (p=0.8284)
SULT1A1 | 215299_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1574)
SULT1A1 | 215299_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3662)
SULT1A1 | 215299_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1029)
SULT1A1 | 215299_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6472)
SULT1A1 | 215299_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SULT1A1 | 215299_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0972
p=0.075
p=0.5863
p=0.3181
p=0.5782
p=0.0348
p=0.5628
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2